

Laya-Bio: Candidate Scoring and Reliability on Short Biological Sequences

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Abstract

We evaluate supervised adaptation of an open candidate-scoring encoder to short biological sequences, with no additional neural continual pretraining. The study separates predictive quality from token compression and candidate-order reliability. Starting from one Laya checkpoint, we train raw-input and biological byte-pair-encoding (BPE) candidate models, a fixed-class readout control, and a text-only control on DNA promoter detection and seven-class protein structural classification. All four conditions use three seeds and a matched supervision budget; final checkpoints and calibration temperatures are fixed before test evaluation. On a common eligible test subset, the biological-BPE candidate model reaches 90.16% DNA accuracy and 59.05% protein accuracy, exceeding the implemented fixed-class control by 1.40 and 7.91 percentage points. However, raw-input candidate scoring is more accurate on both tasks, and biological BPE does not reduce measured training time. Development-set interventions support sequence dependence while exposing candidate-order sensitivity: reordering protein labels changes 9.04% of biological-BPE predictions despite a small aggregate accuracy change. These results support candidate scoring as a workable supervised interface in this setting, while limiting claims about tokenization benefits, order invariance, and generalization beyond the evaluated subsets.

1 Introduction

A candidate-scoring interface takes a sequence, a task description, and a finite set of textual labels, then returns a distribution over those labels. Such an interface can expose several classification tasks through one model without generating free-form answers. Whether it remains accurate and reliable after adaptation to biological sequences is an empirical question: a model can exploit fixed templates, respond to candidate positions, or compress sequences without improving their representation.

We study this question using the open Laya typed-decision encoder (NandhaKishorM, 2026). Our scope is two supervised, closed-set tasks: DNA promoter detection and coarse protein structural classification. We compare raw sequence tokenization with a historical biological BPE vocabulary, and compare candidate scoring with an implemented fixed-class readout and a trained text-only control. The biological vocabulary is a functional input option, not an assumed source of improvement. No additional neural continual pretraining (CPT) is performed.

The test results distinguish interface utility from tokenization gains. With biological BPE, candidate scoring exceeds the implemented fixed head by 1.40 percentage points on DNA accuracy and 7.91 points on protein accuracy, averaged over three seeds. Yet the raw-input candidate model is more accurate than its biological-BPE counterpart on both tasks. Development-set diagnostics further show that candidate reorderings can change individual predictions while leaving aggregate accuracy nearly unchanged.

This study contributes (i) a reproducible adaptation and fixed-checkpoint evaluation protocol, (ii) matched-supervision comparisons of two input representations and two output interfaces, and (iii) diagnostics separating sequence dependence, candidate-order stability, and probability calibration. These are empirical contributions within the specified data and controls. The experiments do not establish unseen-task understanding, superiority to specialized biological encoders, or a new biological foundation model. We first place the study relative to existing sequence-text methods, then describe the protocol and its results.

2 Related work

Sequence–text alignment and instruction tuning. Joint use of biological sequences and natural language predates this study. ProtST augments protein representation learning with biomedical descriptions and evaluates supervised and zero-shot prediction (Xu et al., 2023b). BioTranslator maps biological features and textual descriptions into a shared space for zero-shot biomedical classification (Xu et al., 2023a). These methods motivate explicit attention to label semantics, but our two fixed label sets do not test their zero-shot setting. A candidate string can be present in an input without the model demonstrating transferable semantic understanding.

Instruction-tuned biological models broaden the interface further. LLaMA-Gene combines vocabulary expansion, additional sequence pretraining, and instruction fine-tuning (Wang Liang, 2024). Its instruction corpus is also a source of the two BioPAWS-2 task views used here. ChatNT connects biological-sequence inputs with a conversational language interface (de Almeida et al., 2025). Our experiment evaluates a finite candidate distribution from an encoder rather than open-ended answer generation, and omits additional neural sequence pretraining. The differing data, objectives, and evaluation settings preclude cross-paper accuracy rankings.

Tokenization and encoder adaptation. DNABERT-2 uses BPE in a genome foundation model and evaluates its representation and computational benefits (Zhou et al., 2024). That evidence does not imply that adding a biological vocabulary to a text-derived checkpoint will improve a small supervised adaptation. Our representation comparison holds examples and update counts fixed while recording the larger embedding matrix and measured training cost. Laya builds a typed-decision interface over an encoder; the checkpoint used here employs ModernBERT-large (Warner et al., 2025; NandhaKishorM, 2026). We retain the checkpoint’s 1,024-token configuration rather than infer a longer usable context from the underlying encoder family.

Calibration and evaluation scope. Temperature scaling adjusts confidence after training without changing the most probable class (Guo et al., 2017). We fit temperatures on a separate calibration split and report test probability metrics alongside classification metrics. Sequence interventions and candidate permutations address different questions: the former probe dependence under changed inputs, whereas the latter preserve the candidate set and allow semantic alignment of predictions. Neither substitutes for homology-aware partitioning or evaluation on new task descriptions.

3 Model and adaptation

Candidate scoring. An example comprises task context q , sequence s , ordered candidate strings $C = (c_1, \dots, c_K)$, and target index y . Here $K \in \{2, 7\}$. The original Laya builder places a marker before each candidate and includes the task instructions and sequence state. The encoder, choice-type embedding, and two contextual head blocks produce candidate-marker states h_j . A shared layer-normalization/MLP scorer yields $z_j = g_\theta(h_j)$. We minimize ordinary supervised cross-entropy,

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp z_{i,y_i}}{\sum_{j=1}^{K_i} \exp z_{i,j}}. \quad (1)$$

We train the encoder, input embeddings, contextual head, type embedding, and candidate scorer. The unused action head is frozen. This is a CE adaptation, not a reproduction or comparison of the upstream reinforcement-learning objective. Training candidate order is a deterministic hash-based permutation of each sample ID and seed; it remains fixed for that run rather than changing each epoch. Targets follow the permutation. Standard evaluation uses the canonical order.

Biological BPE. The *raw candidate* condition uses the original Laya tokenizer for the complete textual input. The *BioBPE candidate* condition tokenizes biological sequences separately with historical DNA/protein BPE vocabularies and maps their IDs directly to added model rows. Natural-language context and candidates retain the original tokenizer. The historical source recipe sampled about 1 GiB per modality and used BPE minimum frequency 10. These are externally fitted vocabularies, not vocabularies fitted exclusively on this experiment’s training split.

The source vocabularies omitted DNA N and protein N/J, allowing silent character loss. We append those characters in a copy, preserve existing IDs and merges, and require the concatenated source pieces and inverse model-ID mapping to reconstruct every encoded sequence. Excluding source padding/unknown symbols gives 19,999

Task	Train	Dev	Calibration	Test
DNA promoter (2 classes)	16,766	1,052	1,053	2,145
Protein structure (7 classes)	15,284	926	933	1,952

Table 1: Common eligible examples after data cleaning and the inherited complete-input filter. Dev is used for diagnostics; calibration is used only for temperature fitting.

DNA and 8,000 protein additions: 50,368 original rows become 78,367 rows. For a new piece v decomposed by the original tokenizer as $b_1, \dots, b_m$, initialize

$$E_{\text{new}}(v) = \frac{1}{m} \sum_{r=1}^m E_{\text{old}}(b_r). \quad (2)$$

The decomposition uses the bare biological fragment, without marker text. Original embedding rows remain unchanged at initialization. Added rows and the model are then optimized by the same supervised loss. Some added tokens never occur in the training set, so not every row receives direct sequence supervision.

Controls. The *BioBPE fixed head (B1)* uses the same initial encoder, biological embeddings, contextual head blocks, type embedding, and readout MLP. It omits candidate strings, pools the CLS state, and uses newly initialized two- and seven-class linear output matrices selected by the known task. Its 449.709 million trainable parameters are close to the candidate model’s 449.701 million, but its input and pooling operation differ. The *text-only candidate* retains the candidate architecture and training budget while removing sequences during both training and evaluation. In canonical evaluation, all examples of a task then have identical inputs. This trained control is distinct from removing a sequence only at inference time.

Calibration. For each checkpoint and task, fit a scalar $T \in [0.05, 20]$ by minimizing calibration-set negative log-likelihood (NLL), and evaluate $p_j = \text{softmax}(z/T)_j$ on test. These temperatures are fixed before test inference. Positive T does not change class predictions. Raw and calibrated probability metrics are reported separately.

4 Data and evaluation protocol

Tasks and provenance. We use local BioPAWS-2 JSONL views (Wang, 2026) whose source fields identify LLaMA-Gene DNA/protein instruction data (Wang Liang, 2024). Promoter inputs contain 300 nucleotides; protein inputs contain at most 512 residues and belong to seven broad structural classes. This is structural-class prediction, not fine-grained fold identification. Task context is extracted from the user message, while the assistant answer is used only as a target. Each task has one context template and one fixed canonical candidate set. Local file hashes, rather than a mutable dataset page, identify the exact experimental data.

Cleaning and partitioning. We remove within-split exact duplicates, and exclude train/validation exact-sequence label conflicts. DNA partition groups also use reverse-complement canonicalization; conflicting train/validation groups are excluded without relabeling. Retained training groups take precedence over validation. The remaining validation groups are split approximately equally, stratified by class, into dev and calibration using seed 20260922. Test is deduplicated and checked against retained training/validation groups after those memberships are fixed. Test labels do not determine filtering or partitions. Exact/partition-group overlap is zero; this does not establish homology independence.

Table 1 uses the complete-input eligibility list frozen in an earlier representation audit. That audit used longer wrapped sequence encodings; its filter excludes 309 protein training, 13 dev, 9 calibration, and 42 test examples from the cleaned views. All DNA examples in the cleaned views are retained. We retain this common subset for every final model even though the final representations are shorter. Thus the protein test covers 1,952/1,994 (97.89%) of the cleaned test view, and results are conditional on this inherited subset. No final input is truncated.

Training and fixed test evaluation. Each condition starts independently from the same Laya checkpoint and uses seeds 20260922, 20260923, and 20260924. Joint training samples the combined tasks without task reweighting. Every run uses 3,005 updates, 96,160 example presentations, micro-batch 8 with accumulation 4, AdamW at 2×10^{-5} ,

Model	DNA Acc.	DNA F1	Protein Acc.	Protein F1
Raw candidate	91.10 $\pm$ 0.35	91.09 $\pm$ 0.35	59.97 $\pm$ 0.56	50.63 $\pm$ 1.67
BioBPE candidate	90.16 $\pm$ 0.14	90.16 $\pm$ 0.14	59.05 $\pm$ 0.33	50.28 $\pm$ 0.42
BioBPE fixed head (B1)	88.76 $\pm$ 0.37	88.76 $\pm$ 0.37	51.14 $\pm$ 5.36	38.00 $\pm$ 4.15
Text-only candidate	49.84 $\pm$ 0.57	33.26 $\pm$ 0.25	29.10 $\pm$ 0.00	6.44 $\pm$ 0.00

Table 2: Fixed-checkpoint test results (% , mean $\pm$ sample SD over three seeds). F1 is macro-F1 over the fixed class set. All models use the same eligible examples.

Model	Task	NLL (raw)	NLL (cal.)	Brier (cal.)	ECE (cal.)
Raw candidate	DNA	0.243 $\pm$ 0.014	0.222 $\pm$ 0.010	0.133 $\pm$ 0.007	0.011 $\pm$ 0.005
BioBPE candidate	DNA	0.314 $\pm$ 0.011	0.249 $\pm$ 0.002	0.147 $\pm$ 0.002	0.022 $\pm$ 0.001
BioBPE fixed head (B1)	DNA	0.300 $\pm$ 0.012	0.279 $\pm$ 0.008	0.165 $\pm$ 0.005	0.018 $\pm$ 0.001
Text-only candidate	DNA	0.693 $\pm$ 0.000	0.693 $\pm$ 0.000	0.500 $\pm$ 0.000	0.004 $\pm$ 0.003
Raw candidate	Protein	1.190 $\pm$ 0.034	1.044 $\pm$ 0.024	0.529 $\pm$ 0.009	0.054 $\pm$ 0.011
BioBPE candidate	Protein	1.184 $\pm$ 0.023	1.057 $\pm$ 0.014	0.539 $\pm$ 0.004	0.049 $\pm$ 0.002
BioBPE fixed head (B1)	Protein	1.241 $\pm$ 0.059	1.236 $\pm$ 0.063	0.619 $\pm$ 0.037	0.042 $\pm$ 0.014
Text-only candidate	Protein	1.630 $\pm$ 0.001	1.628 $\pm$ 0.001	0.781 $\pm$ 0.000	0.009 $\pm$ 0.007

Table 3: Test probability metrics (mean $\pm$ sample SD). Calibration uses the previously fitted task/checkpoint temperature. Brier is the sum of squared class-probability errors per example; ECE uses 15 equal-width bins. Lower values are better.

weight decay 0.01, and 5% warmup followed by cosine decay. We use BF16 on one RTX 4080 SUPER and select the last checkpoint, without early stopping. This matches supervision, not FLOPs or parameter count across representations.

Before test inference, a manifest fixes all twelve checkpoints, their temperatures, input files, and evaluation code. Each checkpoint must reproduce its full saved dev logits before evaluation. Test is evaluated once per checkpoint at batch size 16, with no retraining, seed selection, or temperature refitting. We report task-specific accuracy and macro-F1, with balanced accuracy and Matthews correlation coefficient (MCC) in the appendix. Probability metrics are NLL, multiclass Brier score, and expected calibration error with 15 equal-width confidence bins (ECE15). Means and sample standard deviations summarize the three training seeds; they are not confidence intervals over independent biological samples.

5 Results and diagnostics

Candidate scoring versus the implemented fixed head. The BioBPE candidate model attains 90.16% DNA accuracy and 59.05% protein accuracy (Table 2). Relative to B1, the paired mean differences are $+1.40 \pm 0.42$ and $+7.91 \pm 5.69$ percentage points, respectively; all three seeds favor candidate scoring on both tasks. Protein macro-F1 increases by 12.28 ± 4.56 points. B1 protein accuracy varies substantially, from 45.65% to 56.35%. This comparison supports the current candidate configuration under the tested budget, but input format, pooling, and output-layer initialization differ, so it does not isolate a causal benefit of candidate semantics.

Token compression does not yield a predictive or training-time gain. Raw candidate scoring has higher test accuracy than BioBPE on both tasks: the paired BioBPE differences are -0.93 ± 0.29 points for DNA and -0.92 ± 0.49 points for protein, negative in all seeds. The dev DNA difference was $+0.60$ points, so that advantage does not carry to test. The pooled training-input median falls from 173 to 96 tokens, but mean training time is 41.20 minutes for raw and 41.24 minutes for BioBPE. Peak allocated memory rises from 7.89 to 8.41 GiB. We therefore report compression, a larger embedding matrix, and observed cost rather than infer acceleration from token counts. We did not optimize kernels or measure a separate inference-latency benchmark.

Probability quality. Previously fitted temperatures lower mean BioBPE test NLL from approximately 0.31 to 0.25 for DNA and from 1.18 to 1.06 for protein (Table 3). Raw candidate scoring retains lower calibrated NLL, about 0.22 and 1.04. Calibration gains therefore do not reverse the representation comparison, and NLL improvement alone does not establish perfect calibration. Brier and ECE report complementary aspects of the resulting distributions. The text-only model’s low ECE despite weak classification also shows why ECE alone cannot rank overall predictive quality.

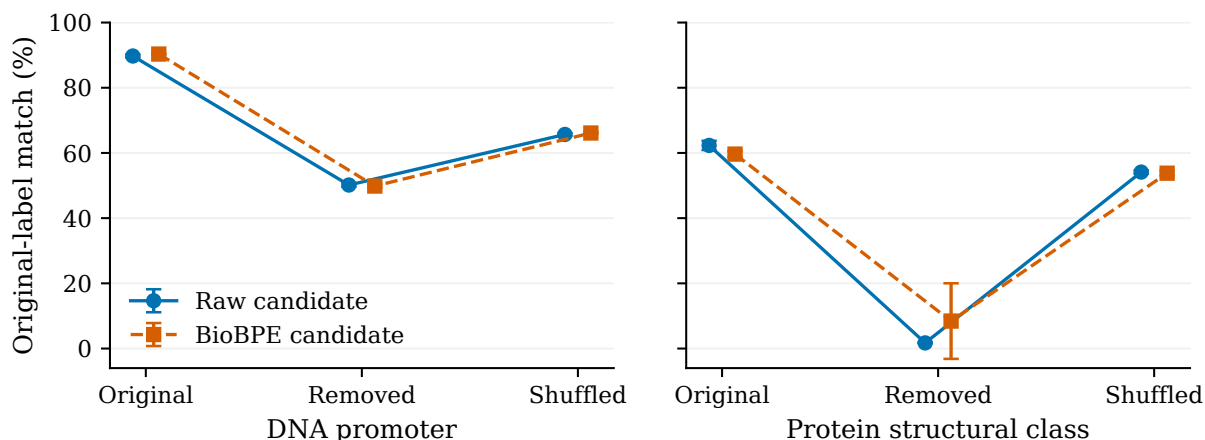


Figure 1: Dev sequence-dependence diagnostics. Points and error bars are means and sample SDs across three trained seeds. For shuffled sequences, each seed first averages three deterministic character permutations. The vertical axis measures agreement with the *original* labels: removed or shuffled sequences are interventions, not independently labeled biological examples.

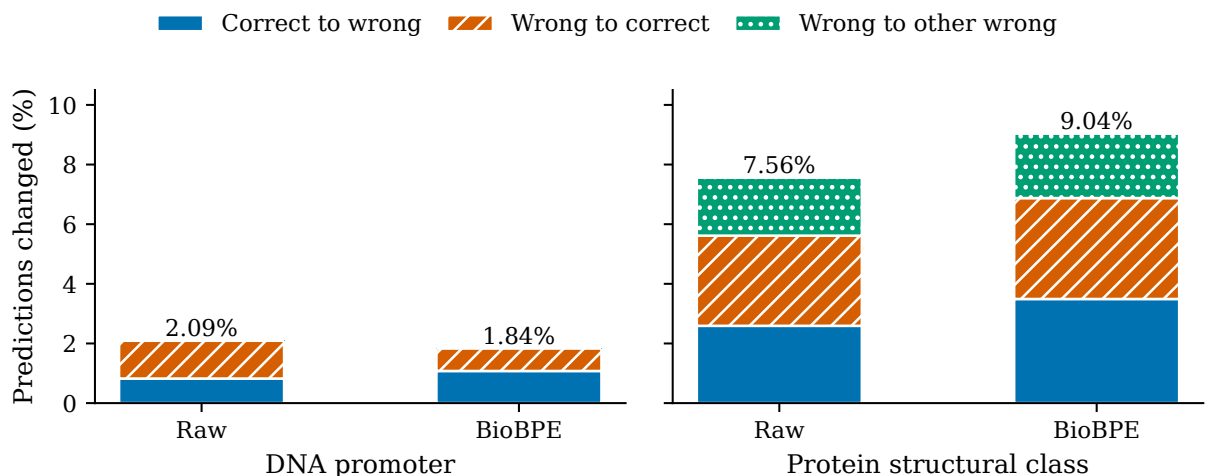


Figure 2: Dev prediction changes after one deterministic nonidentity candidate permutation per example, with outputs mapped back to canonical labels. Stacks show mean fractions over three seeds. For BioBPE protein predictions, correct-to-wrong and wrong-to-correct changes nearly cancel in accuracy, while wrong-to-other-wrong changes also contribute to instability.

Sequence dependence. The trained text-only control reaches 49.84% DNA and 29.10% protein test accuracy, with a constant class prediction within each task/seed. Its low protein macro-F1 reflects the class imbalance. Dev interventions on the sequence-trained BioBPE model reduce original-label matching from 90.34% to 49.84% after DNA removal and to 66.12% after DNA shuffling; protein values change from 59.65% to 8.42% and 53.79%, respectively (Figure 1). Shuffling preserves character counts but not biological labels. These observations support sequence dependence and residual composition-related signal, without identifying a learned biological mechanism. Inference-time removal is also a distribution shift and must not be equated with the separately trained text-only baseline.

Candidate-order sensitivity. The same candidate set does not ensure the same prediction. On dev, reordering candidates changes 1.84% of BioBPE DNA predictions and 9.04% of protein predictions. For protein, 3.49% of examples change from correct to wrong, 3.38% from wrong to correct, and 2.16% between wrong classes (Figure 2; displayed components are rounded). Aggregate accuracy changes by only -0.11 percentage points. Raw protein predictions also show sensitivity, with 7.56% changing. Training with fixed per-example candidate permutations has therefore not produced order invariance. This diagnostic covers one nonidentity permutation per example, not all possible orders, and was not expanded on test.

6 Limitations

The evidence concerns two closed-set tasks, one checkpoint, three seeds, and one implemented fixed-head control. Each task has a fixed template and candidate set; no unseen-task, paraphrase, or new-label evaluation was performed. Candidate scoring’s measured advantage cannot be assigned solely to textual label semantics because the control changes pooling and input structure. Specialized biological encoders and generative sequence–text systems were not evaluated under a matched budget.

Data independence remains limited. Exact and DNA reverse-complement overlap checks do not establish homology-separated protein evaluation or species-separated DNA evaluation. Historical cluster summaries lack a verified membership mapping to these local records. The biological BPE was fitted on external corpora whose overlap with the benchmark is not fully established, and the parent checkpoint’s historical exposure is not excluded. The inherited eligibility filter also restricts coverage. These limitations rule out an unqualified leakage-free generalization claim.

This was an iterative development study, not a publicly preregistered experiment. Earlier compact-vocabulary runs informed development; the final twelve-model protocol was fixed before its test predictions. Seed SDs do not quantify biological sampling uncertainty, and no significance claim is made. The observed order sensitivity and incomplete probability calibration limit direct deployment claims. The study is a computational benchmark, with no experimental or clinical validation.

7 Conclusion

Under a fixed supervised budget, a biological-BPE Laya candidate model outperforms the implemented fixed-class and text-only controls on the evaluated short-sequence tasks. Raw candidate input nevertheless gives higher test accuracy, and biological BPE does not improve measured training time. Development diagnostics show that the models use sequence information while remaining sensitive to candidate order. The resulting evidence supports a bounded adaptation baseline and a reliability protocol, rather than a general tokenization benefit or open-ended biological reasoning capability. Further method development requires evaluation designs that address homology and template transfer without reusing this test set for selection.

The data, code, final checkpoints, and historical corpus archive are publicly available at the fixed revisions listed in Appendix A.

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A Reproducibility and complete results

Versions and artifacts. The starting model is `convaiinnovations/laya-typed-decisions`, revision `f9ab0b228f0fc0f14d873dbc99038f135c2da1b2`. The upstream Laya source revision is `573e5b62696ba441230cd6be71d593331b5d23af`. The initial weight SHA-256 is `4fa56de72383a9d3efa9cfa78955733c81b9fc8067a587ca4beb82c78107a24e`. The final test manifest hashes 288 files and is stored at `artifacts/laya_locked_test/manifest.json`. Its SHA-256 is `9b1594ec0255318ff30486fbea20e3a0172ca1292d1661c040c67800ac86c10e`. The manifest predates test inference and is checked again after all evaluations. Every checkpoint exactly reproduced its complete dev logits; all test logits were finite. Each model evaluated the same 4,097 sample IDs without truncation. Independent sklearn/SciPy metric recomputation agreed to a maximum absolute error of 2.22×10^{-16} .

Data, code, and model availability. The benchmark snapshots, cleaned splits, paper-eligible subsets, frozen manifests, per-example predictions and logits, training/evaluation code, BPE assets, and manuscript materials are publicly available at `dnagpt/laya-bio` (revision `0307d5910b0a37f54caecd444241db6b56331701`). The default configuration contains 32,050 training, 1,978 selection-dev, 1,986 calibration, and 4,097 test examples; the larger cleaned data are provided separately and are not the paper’s evaluation subset.

All twelve final checkpoints (four conditions, three seeds), the original initialization weights, tokenizers, configurations, and loading code are available at `dnagpt/laya-bio-models` (revision `6d727dd4125d783ab719a35c6b55dd7a69276a8d`; 22.46 GB). Seven historical corpus files and the two actual BPE-training samples are archived at `dnagpt/laya-bio-historical-corpora` (revision `bd70a7d157741f8d19ad150adfa350a9b02ba417`). This archive preserves 106.98 GB of original bytes in 404 lossless Zstandard shards; the 42.55 GB release includes manifests and a SHA-256-verifying restoration script. These historical resources do not imply additional neural continual pretraining in this study.

Release files were checked against remote sizes and content hashes. Anonymous access was verified, including a complete download and restoration of the archived PDB 3Di file. Source declarations and component-specific license terms are preserved; the archive is not assigned a blanket license. Upstream provenance and benchmark overlap remain incompletely resolved, as discussed in the limitations. The released local snapshots, rather than the current BioPAWS-2 page contents, define the experimental data.

Implementation details. Input length is capped at 1,024 tokens and the upstream candidate header budget is 256. The original builder’s choice instruction is *Choose the correct biological label for the sequence*. The state prefix contains **Task context:** followed by the cleaned task instruction and **Sequence:** followed by the sequence. Candidate strings are the canonical class names in Table 9; gold answers are not part of the input. The protein class labels are reproduced as dataset categories, without treating them as a complete structural ontology.

The joint training loop shuffles the combined sample list without class weighting. It uses gradient clipping at norm 1.0, 151 warmup updates, and the final fixed-budget checkpoint. The budget is approximately three passes, but example presentations are reported explicitly because the sampler reshuffles when a full micro-batch no longer fits. AdamW and BF16 settings are shared. Calibration optimizes a single task-specific temperature per checkpoint on the calibration split. Macro metrics retain all two or seven classes; Brier uses the unnormalized sum across classes. ECE15 uses the maximum predicted probability and equal-width bins, including confidence 1 in the final bin.

The historical BPE recipe sampled about 1 GiB each from DNA and protein text using seed 42, with minimum frequency 10. The original vocabulary files, repair details, source scripts, and corpus hashes are preserved. Training observes 18,448 of the 19,999 added DNA tokens and 7,972 of the 8,000 protein tokens. Tokens unobserved in sequences are not claimed to have direct task supervision. The original vocabulary was not retrained on the benchmark.

Development history and test boundary. An earlier single-seed experiment used a compact 64-token extension and wrapped sequence pieces, with a frozen-model reference. Those scores are development history and are not mixed into the final raw-input or BioBPE test comparisons. The complete-input list inherited from that stage was held fixed in the later three-seed study. Subsequent experiments used the complete historical BPE and added B1/text-only controls and dev sequence/order diagnostics. The final test included only the four conditions and three seeds listed in Table 4. No test-dependent perturbation, hyperparameter search, or new training followed this evaluation.

Seed	Model	DNA Acc.	DNA F1	Protein Acc.	Protein F1
20260922	Raw candidate	91.14	91.14	60.55	52.56
20260922	BioBPE candidate	90.30	90.30	59.38	50.70
20260922	BioBPE fixed head (B1)	88.48	88.48	45.65	33.44
20260922	Text-only candidate	49.51	33.12	29.10	6.44
20260923	Raw candidate	91.42	91.42	59.94	49.72
20260923	BioBPE candidate	90.16	90.16	58.71	49.86
20260923	BioBPE fixed head (B1)	89.18	89.18	56.35	41.54
20260923	Text-only candidate	50.49	33.55	29.10	6.44
20260924	Raw candidate	90.72	90.72	59.43	49.61
20260924	BioBPE candidate	90.02	90.02	59.07	50.28
20260924	BioBPE fixed head (B1)	88.62	88.62	51.43	39.02
20260924	Text-only candidate	49.51	33.12	29.10	6.44

Table 4: All test seeds; no best-seed selection. Classification metrics are percentages.

Model	Task	Bal. Acc.	MCC	Brier (raw)	ECE (raw)
Raw candidate	DNA	91.10 $\pm$ 0.35	0.822 $\pm$ 0.007	0.138 $\pm$ 0.008	0.036 $\pm$ 0.005
BioBPE candidate	DNA	90.16 $\pm$ 0.14	0.803 $\pm$ 0.003	0.158 $\pm$ 0.002	0.060 $\pm$ 0.003
BioBPE fixed head (B1)	DNA	88.76 $\pm$ 0.37	0.775 $\pm$ 0.007	0.171 $\pm$ 0.002	0.043 $\pm$ 0.014
Text-only candidate	DNA	50.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.500 $\pm$ 0.000	0.006 $\pm$ 0.005
Raw candidate	Protein	49.40 $\pm$ 1.81	0.483 $\pm$ 0.007	0.574 $\pm$ 0.011	0.181 $\pm$ 0.000
BioBPE candidate	Protein	49.12 $\pm$ 0.81	0.471 $\pm$ 0.004	0.579 $\pm$ 0.003	0.165 $\pm$ 0.005
BioBPE fixed head (B1)	Protein	39.50 $\pm$ 4.29	0.370 $\pm$ 0.074	0.620 $\pm$ 0.036	0.049 $\pm$ 0.017
Text-only candidate	Protein	14.29 $\pm$ 0.00	0.000 $\pm$ 0.000	0.781 $\pm$ 0.000	0.008 $\pm$ 0.001

Table 5: Additional test metrics, mean $\pm$ sample SD. Balanced accuracy is in percent; MCC, Brier, and ECE are unscaled. Calibrated Brier and ECE are in Table 3.

Model	Trainable (M)	Training (min)	Peak allocated (GiB)
Raw candidate	421.030	41.20 $\pm$ 0.90	7.89 $\pm$ 0.00
BioBPE candidate	449.701	41.24 $\pm$ 0.82	8.41 $\pm$ 0.00
BioBPE fixed head (B1)	449.709	41.71 $\pm$ 1.01	8.41 $\pm$ 0.00
Text-only candidate	449.701	40.63 $\pm$ 0.60	8.40 $\pm$ 0.00

Table 6: Measured training cost and trainable parameter counts. Times exclude subsequent evaluation and checkpoint reload checks. Peak allocated memory is the PyTorch training measurement, not the GPU driver’s full process footprint. Equal supervision does not imply equal computation.

Task	Representation	Train median	Train p95	Train max
DNA	Raw candidate	174	180	194
DNA	BioBPE candidate	93	97	103
Protein	Raw candidate	145	258	318
Protein	BioBPE candidate	127	216	282

Table 7: Full training-input lengths, including task context and candidates, on the common subset. Values are tokens, not nucleotides or residues.

Model	Task	Original	Removed	Shuffled	Order agreement
Raw candidate	DNA	89.73 $\pm$ 0.48	50.16 $\pm$ 0.55	65.67 $\pm$ 0.45	97.91 $\pm$ 0.72
Raw candidate	Protein	62.31 $\pm$ 1.41	1.73 $\pm$ 0.00	54.13 $\pm$ 0.54	92.44 $\pm$ 0.19
BioBPE candidate	DNA	90.34 $\pm$ 0.05	49.84 $\pm$ 0.55	66.12 $\pm$ 0.31	98.16 $\pm$ 0.62
BioBPE candidate	Protein	59.65 $\pm$ 0.87	8.42 $\pm$ 11.60	53.79 $\pm$ 0.64	90.96 $\pm$ 0.51

Table 8: Dev diagnostics (%). Original is accuracy; removed/shuffled are matching rates against original labels, without assuming intervention-label validity. Order agreement compares semantic predictions after canonicalizing the permutation. Values are mean $\pm$ sample SD.

Task	ID	Canonical class	Test n
DNA	0	Non-promoter	1062
DNA	1	promoter	1083
Protein	0	All Alpha	328
Protein	1	All Beta	430
Protein	2	Alpha and Beta	568
Protein	3	Alpha plus Beta	456
Protein	4	Multi-domain Proteins	48
Protein	5	Mixed Structures	19
Protein	6	Small Proteins and Peptides	103

Table 9: Canonical task classes and eligible test support. The trained text-only protein control predicts class 2 for every example in all three seeds.